TerraTrace: Reproducible, Uncertainty-Aware, Multi-Lens Morphological Analog Screening Across Five Worlds

NASA Space Apps 2026 - research preview. Repository: `github.com/pal-123456789/terratrace`

> **Scope / claim firewall.** TerraTrace screens terrain *morphology* between curated map products. It does **not** estimate rover traversability, validate autonomy, establish geological equivalence, certify site access, or certify mission readiness. Earth sites named here are *illustrative analogs*, not endorsed field sites. It is not NASA-endorsed.

Abstract

Choosing a terrestrial analog for a planetary terrain is usually done by expert judgment and reported as a single qualitative verdict. TerraTrace makes that screening **quantitative, reproducible, and honest**. It brings a planetary reference terrain and candidate Earth terrains to a common metric support on each body's own geometry, computes a battery of map-scale morphology descriptors, and ranks candidates by **Pareto dominance across several independent descriptor "lenses"** - deliberately refusing to collapse the comparison into one composite "likeness" score. When lenses disagree it returns an explicit TIE / INDETERMINATE state instead of inventing a winner, and attaches a spatial block-bootstrap, a permutation significance test, a non-composite decision-confidence verdict, and a full provenance record to every result. In exploratory runs on real data (Mars MOLA vs three Copernicus GLO-30 Earth terrains) the method favoured a documented basalt analog - Craters of the Moon, Idaho - on the core roughness/relief axes. A **pre-registered** follow-up then tested that preference under matched physical footprints, supports at or above MOLA's effective resolution, three held-out Martian volcanic plains, and two contrast terrains. **The preference did not survive:** in the honest regime the engine abstained for all seven Mars references, and Craters was the nearest candidate for a cratered highland as readily as for volcanic plains - the signature of a non-specific "least-dissimilar member of a weak shortlist" effect rather than a specific analog. TerraTrace's contribution is therefore a screening method that **detects its own false analogs** with rules fixed before the data are seen. Everything is offline-reproducible from one command and covered by 771 automated tests.

1. Problem and approach

Analog site selection underpins instrument testing, operational rehearsal, and the interpretation of orbital data. Yet "X is a good analog for Y" is rarely stated with a defined support scale, an uncertainty, or a falsifiable procedure. TerraTrace treats analog screening as a transparent, auditable computation with three commitments: (i) **comparison at a declared common support** on each body's correct geometry; (ii) **no composite score** - candidates are ranked by Pareto dominance over interpretable axes, and ties are reported, not broken; and (iii) **every number carries provenance and uncertainty**. The goal is not to declare a winner but to say, defensibly, *what transfers, at what scale, and with what confidence.*

2. Method

2.1 Common support and per-body reprojection. Each area of interest is warped to its own local azimuthal-equidistant metric grid using the correct body radius (Mars 3,396,190 m; Moon 1,737,400 m; Mercury 2,439,400 m; Ceres 469,730 m; Earth WGS84) and aggregated to a declared common support (e.g. 1000 m), so descriptors compare like with like rather than conflating pixel size with terrain.

2.2 Morphology descriptors. Core: slope, local relief, detrended ruggedness, low-slope connectivity. Extended: aspect dispersion, hypsometric integral. Geomorphic: Zevenbergen-Thorne curvature and the Vector Ruggedness Measure. Landform: geomorphon landform composition and topographic openness. Scale & fabric: the radially-averaged power-spectral slope and a directional-structure-function anisotropy (fabric strength and orientation). Planetary roughness: the Kreslavsky-Head (2000) differential-slope baseline roughness (the metric behind NASA's global MOLA roughness maps) and the Iwahashi-Pike (2007) three-signature (slope / convexity / texture) nested-means terrain class. Dissection: a depression-filled D8 flow-accumulation surface-dissection texture (a purely geometric routing descriptor). The bank also carries a Haar wavelet multiscale-energy spectrum, the Strahler hypsometric curve, surface-normal-dispersion roughness, and a Weiss (2001) TPI slope-position classification. A Topographic Wetness Index (Beven & Kirkby 1979) and a convergence index (Kothe & Lehmeier 1996) summarise flow-geometric convergence; a sky-view factor (Zaksek et al. 2011) captures hemispherical openness;

and a two-scale TPI slope-position classification (Weiss 2001) resolves nested landform position. Finally a differential box-counting fractal dimension, gliding-box lacunarity, Shannon terrain entropy, and a Flint (1974) slope-vs-contributing-area concavity complete the battery. A geostatistical semivariogram (range/sill/nugget; Matheron 1963) and an RMS/Allan-deviation roughness length (Shepard et al. 2001) summarise spatial structure; Wood (1996) morphometric curvature classes and a Woodcock (1977) surface-normal orientation-tensor fabric add shape descriptors; an openness-difference slope position (Yokoyama 2002) and the Melton (1965) ruggedness number complete the batch. Distributions are compared to the reference with scale-normalized Wasserstein/KS distances.

2.3 Decision: Pareto, not a score. Candidates become vectors of distances-to-reference and are ranked by **Pareto dominance** (peeled non-dominated layers + a dominance matrix). The result is one of `RECOMMEND_FOLLOW_UP`, `TIE`, `INDETERMINATE`, `NOT_COMPARABLE`, or `BLOCKED`. Six **lenses** - core, extended, geomorphic, landform, texture, hydro - are independent axis sets (the hydro lens adds a Topographic Wetness Index, aspect convergence, and flow-dissection distance-to-reference axes); a recommendation that holds under more lenses is more robust, and lens disagreement is surfaced, not hidden.

2.4 Uncertainty, significance, confidence. A spatial moving-block bootstrap gives autocorrelation-aware distance intervals and a non-domination frequency; a one-sided permutation test asks whether the recommended unit is genuinely closer to the reference than the runner-up; and a **non-composite decision-confidence** verdict (`robust` / `qualified` / `fragile` / …) restates the bootstrap, significance, and lens-stability evidence as three independent signals, explicitly naming any signal it could not evaluate.

2.5 Provenance and reproducibility. Every study is hash-anchored (sha256), carries an evidence graph and a transfer-gap ledger, and exports a self-contained bundle (study package, Markdown + offline-HTML report, viewer, references, checksums, confidence). A fixed synthetic study is byte-reproducible within an environment; portable invariants (not pinned digests) gate it across platforms. The full geomorphometric descriptor bank and a swath/profile `transect` extractor are reachable from the unified CLI (`bank`, `transect`) and the loopback-only HTTP API (`POST /bank`, `POST /transect`, the same host-allowlist and body-cap guards as the existing endpoints), and the terrain fingerprint carries an opt-in geostatistical similarity term (`include_geostat`).

3. Data

Mars reference: MGS **MOLA 463 m** (USGS Astrogeology, keyless). Earth candidates: **Copernicus GLO-30** via OpenTopography. The engine's source catalog carries 13 products across the five bodies with coverage, licence, and 1-sigma vertical-accuracy metadata; a no-network preflight validates each bounding box, selects the covering product, and sizes the request before any download.

4. Results: an exploratory screen, then a pre-registered test

4.1 Exploratory screen (not controlled - retained as history)

Mars MOLA versus three real Copernicus GLO-30 Earth terrains at 1000 m support. Distance to the Mars reference per axis (lower = closer):

candidate	slope	relief	ruggedness	connectivity	aspect	hypsometric
Craters of the Moon, ID	**0.75**	**0.78**	**0.76**	**0.018**	0.39	0.28
Nevada basin	1.33	1.38	1.45	0.030	**0.011**	**0.14**
Andean volcano	1.73	1.50	1.85	0.057	0.012	0.17

Four-lens ablation on the **same** data (one command, `--ablation`):

lens	result	recommended	non-dominated set
core	RECOMMEND	**Craters of the Moon**	{Craters}
extended	TIE	-	{Craters, Nevada}
geomorphic	RECOMMEND	**Craters of the Moon**	{Craters}
landform	TIE	-	{Craters, Nevada, Andes}

Exploratory reading (superseded by 4.2). On the core and geomorphic lenses Craters of the Moon was the closest of the three candidates, which looked like an independent re-identification of a known basalt analog; the extended and landform lenses tied. Running the one-command atlas over four Mars terrains (Jezero, Gale, Daedalia Planum, Elysium Planitia) kept Craters non-dominated in all 16 referencexlens cells. These runs, however, compared a ~10 deg Mars window with ~0.4 deg Earth windows and used supports below MOLA's effective resolution, so they were treated as a **hypothesis to test**, not a result.

4.2 Pre-registered matched-footprint test (protocol v2)

The hypothesis was tested under a protocol frozen and committed before the runner existed (`docs/BENCHMARK-PROTOCOL-v2.md`): every unit cropped to the same untruncated 24 km window; evidence only at supports >= MOLA's effective resolution (1.5 and 2 km); three held-out Martian volcanic plains (Amazonis, Syria, Hesperia) as confirmation; two contrast terrains (Noachis Terra highland, Coprates Chasma canyon) as a specificity control; a resolution-matching control; and an absolute-adequacy check against each reference's own variability.

hypothesis	frozen verdict
H1 - held-out plains recommend Craters	**not supported** (0 of 3)
H2 - Craters not chosen for contrast terrains	inconclusive (no honest-regime recommendation anywhere)
H3 - not a source-smoothing artifact	not supported (primary changed under resolution matching)
H4 - absolute adequacy	reported (no recommended unit)

Reading. In the honest regime the engine abstained for all seven references. Craters was nonetheless the nearest candidate in 11 of 14 cells - **including the cratered highland Noachis Terra** - and under resolution matching it became the stable recommendation in those same cells, with the engine's own specificity verdict `non_specific`. It also fell outside the own variability of Daedalia and Elysium, the two plains the exploratory claim was built on. The most economical explanation is a shortlist effect: at kilometre scale Martian terrain is smoother than all three Earth tiles, and Craters is simply the smoothest of them. The exploratory analog claim is therefore **withdrawn**; full results, including the non-pre-registered descriptive analyses and a protocol-design lesson, are in `docs/RESULTS-v2.md`. A direct test of the shortlist explanation (smoothness-matched non-volcanic controls, wider windows for power) is proposed as protocol v3.

5. Validation

Correctness is pinned by a scientific-correctness suite of 51 oracles that drive the real engine on synthetic terrains with known answers: self-distance is zero; slope/relief/ruggedness are invariant to a constant offset and to 90 deg rotation; roughness discrimination is monotone; a planted close twin is recovered over an impostor; a water tile is BLOCKED; openness is 90 deg on a plane and exceeds 90 deg at a peak; geomorphons label a cone apex a peak and a basin centre a pit; a sinusoid's spectral peak lands at its wavelength; white noise is spectrally flat (Parseval holds); a ridged surface reports the correct fabric orientation. Separate synthetic ground truths validate the comparison controls themselves: a footprint mismatch that flips the recommended analog is detected and corrected; a source-smoothing false analog is removed by resolution matching; and a non-specific "wins everywhere" candidate is distinguished from a specific one. Passing these tests establishes software and method correctness, not a scientific discovery - the real-data hypothesis test is reported separately in §4.2.

6. Limitations and transfer gaps (stated, not hidden)

TerraTrace compares *map-scale morphology of sampled DEMs* and nothing else. It says nothing about composition, rock-scale obstacles below the pixel, trafficability, or mission suitability; those bounds are reproduced verbatim in every report. Two concrete limits of the exploratory example (§4.1): the Mars window (~10 deg) and the Earth windows (~0.4 deg) differed in footprint by ~15x, and supports below MOLA's effective resolution compared interpolated Mars detail with real Earth detail - both corrected in the pre-registered test of §4.2. That test is itself limited: 12-16 cells per side at 24 km, one source per body (not independent replications), a 3x effective- resolution heuristic rather than a measured MOLA transfer function, and only three Earth candidates. The independent-pixel bootstrap is optimistic relative to the spatial block bootstrap on autocorrelated terrain. Deferred work:

DEM co-registration and vertical-datum error propagation end-to-end into the decision; wider Earth windows and smoothness-matched controls (protocol v3).

7. Reproducibility

```
pip install -r terratrace/requirements.txt      # core: NumPy only
python terratrace/tests/run_tests.py             # 771 offline tests
python terratrace/scripts/run_validation.py      # 51 correctness oracles
# real five-world study, one command (needs a free OpenTopography key):
python terratrace/scripts/run_real_analog_study.py --config config/real_multibody.example.yml
```

CI runs the suite and the validation oracles on Python 3.10/3.11/3.12. Large DEMs and generated bundles are git-ignored; everything else regenerates deterministically.

References

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Morphology screening and resampling statistics only - not a traversability, geologic-equivalence, site-access, or mission-readiness claim.